

UChicago REU Notes Week 2

Zih-Yu Hsieh

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1 Week 2

1.1 Intro

For this portion, I want to start with some basic definitions covered by professor May in his office, about the definition of spectra / prespectra / generalized cohomology theory. I'll probably skip some of the details in the intro to make sure I can keep on track. (Also, I probably involve quite a lot of errors in between...I'll try to fix them)

1.1.1 Brown Representability Theorem, and Generalized Cohomology

Let $h\mathcal{C}$ denotes the homotopy category of based spaces (that admits a CW structure, as on them weak equivalences and homotopy equivalence are the same), then a functor $F : h\mathcal{C}^{\text{op}} \rightarrow \mathbf{Ab}$ is *representable*, if there exists a based space $Y \in h\mathcal{C}$, such that $F \cong [-, Y]$.

Example 1.1. The *reduced cohomology functor* $\tilde{H}^n(-; A)$ is one, for all $n \geq 0$ and a fixed abelian group A (as coefficient). Let $K(A, n)$ be the *Eilenberg-MacLane Space* of abelian group A and index $n \geq 0$ (i.e. the n th homotopy group $\pi_n(K(A, n)) = A$), then the following group isomorphism holds:

$$\tilde{H}^n(X; A) \cong [X, K(A, n)]$$

Using more adjunction property, since $\pi_n(A, n) = A = \pi_{n+1}(K(A, n+1)) = [S^{n+1}, K(A, n+1)] = [S^n \wedge S^1, K(A, n+1)] \cong [S^n, \Omega K(A, n+1)] = \pi_n(\Omega K(A, n+1))$, there is a homotopy equivalence $K(A, n) \xrightarrow{\sim} \Omega K(A, n+1)$.

(Yeah, need to understand more about the details)

So, the regular cohomology we see on spaces, are one type of representable functors.

The classification of all such functors is given by *Brown Representability Theorem*:

Theorem 1.2. Suppose the functor $F : h\mathcal{C}^{\text{op}} \rightarrow \mathbf{Set}$ satisfies:

- $F(\bigvee_{\alpha} X_{\alpha}) \cong \prod_{\alpha} F(X_{\alpha})$ (i.e. maps coproduct to product).
- F maps homotopy pushouts (I believe this is the reduced mapping cylinder?) to weak pullbacks (omit the uniqueness of maps). This condition is equivalent to Mayer-Vietoris axiom (which seems like a weaker form of the gluing axiom in sheaf theory, and restrict to CW subcomplexes instead of open sets).

Then, F is representable by some CW complex C , or $F \cong [-, C]$.

The converse for the theorem also holds, as $[-, C]$ also satisfies the above axioms (need more details here).

Here is a definition of generalized cohomology:

Definition 1.3. A *generalized cohomology*, is a sequence of spaces $\{E_n\}_{n \geq 0}$, with homotopy equivalences $\{\epsilon_n : E_n \xrightarrow{\sim} \Omega E_{n+1}\}_{n \geq 0}$.

In a more explicit form, $E^n(X) := [X, E_n]$ has all $n \geq 0$ satisfies $E^n(X) = [X, E_n] \cong [X, \Omega E_{n+1}]$, which is naturally a group (and an abelian group for $n \geq 2$, due to the abelian group structure on $[X, \Omega^m Y]$ for $m \geq 2$).

I remembered the motivation is that, by Brown Representability Theorem, if a sequence of representable functors $E^* : h\mathcal{C}^{\text{op}} \rightarrow \mathbf{Set}$ satisfy similar properties to ordinary cohomology (for instance, the connecting morphism $E^n \rightarrow E^{n+1}$, and excision axiom?), then Brown Representability Theorem guarantees it must be the above form. So, they're precisely all the generalized cohomology.

1.1.2 Ω -Prespectra, Prespectra, and Spectra

When talking about generalized cohomology, we have a sequence of based spaces $\{E_n\}_{n \geq 0}$, together with homotopy equivalences $\epsilon_n : E_n \xrightarrow{\sim} \Omega E_{n+1}$, which is a prototype of what's called *Ω -Prespectra*.

In general, fixing a based space $X \in h\mathcal{C}$, the sequence $\{\Sigma^n X\}_{n \geq 0}$ has a collection of natural maps $\{\text{id}_n : \Sigma(\Sigma^n X) \rightarrow \Sigma^{n+1} X\}_{n \geq 0}$, which the adjunction (Σ, Ω) guarantees a collection of maps $\{\eta_n : \Sigma^n X \rightarrow \Omega(\Sigma^{n+1} X)\}_{n \geq 0}$. In general these maps are not homotopy equivalence, so these type of sequences are a type of *Prespectra*, called *suspension prespectra*. More generally, *Prespectra* is a sequence of spaces $\{X_n\}_{n \geq 0}$, with morphisms $\{\mu_n : X_n \rightarrow \Omega X_{n+1}\}_{n \geq 0}$.

For the strongest form possible, if such map $\epsilon_n : E_n \rightarrow \Omega E_{n+1}$ is a homeomorphism, then these are called *Spectra*.

Professor May restricts to three specific categories:

- Let $\mathcal{P}$ denotes the *category of prespectra*.
- Let $\mathcal{Q}$ denotes the *category of inclusion prespectra*, meaning each defining morphism $\mu_n : X_n \rightarrow \Omega X_{n+1}$ is an inclusion.
- Let $\mathcal{S}$ denotes the *category of spectra*.

For each category, we require the morphism $\phi : X. \rightarrow T.$ to be a sequence of based maps $\phi_n : X_n \rightarrow T_n$, such that the following diagram commutes:

$$\begin{array}{ccc} \Sigma X_n & \xrightarrow{\Sigma \phi_n} & \Sigma T_n \\ \mu'_{X_n} \downarrow & & \downarrow \mu'_{T_n} \\ X_{n+1} & \xrightarrow[\phi_{n+1}]{} & T_{n+1} \end{array}$$

where the map $\Sigma X_n \rightarrow X_{n+1}$ is based on the adjoint property $[\Sigma X_n, X_{n+1}] \cong [X_n, \Omega X_{n+1}]$. Here professor May had talked about how the prototype of the morphism only requires it to commute up to homotopy, but turns out it's too weak as a condition.

With these morphisms, there is an obvious forgetful functor $\mathcal{S} \rightarrow \mathcal{Q} \rightarrow \mathcal{P}$. It seems to satisfy all the conditions for Freyd's Adjoint Functor Theorem to hold, both of the forgetful functors have a left adjoint.

Here, the left adjoint of $\mathcal{Q} \rightarrow \mathcal{P}$ is hard to construct, yet there is a clear construction for the left adjoint of $\mathcal{S} \rightarrow \mathcal{Q}$, denoted as $L : \mathcal{Q} \rightarrow \mathcal{S}$, by $T. \mapsto (LT).$, where $(LT)_n := \text{colim}_q \Omega^{n+q} T_q$ (due to the inclusion $T_q \rightarrow \Omega T_{q+1}$, it induces $\Omega^{n+q} T_q \rightarrow \Omega^{n+(q+1)} T_{q+1}$, hence having a sequence of direct system).

Professor May had claimed the homeomorphism $(LT)_n \cong \Omega(LT)_{n+1}$ due to the fact that Ω commutes with colimit (I do need to work out the detail, as normally Ω as a right adjoint of Σ would be thought of preserving limits instead of colimits).

1.2 Review: Adjoint Functors

This part is converging to the preparation of the theory of monads :) (as professor May had mentioned it quite often as an essential tool to understand operad, so I decided to do related review). The main reference is Riehl's *Category Theory in Context*.

Definition 1.4. Let $\mathcal{C}, \mathcal{D}$ be two categories. Two opposite functors $F : \mathcal{C} \rightarrow \mathcal{D}, \mathcal{C} \leftarrow \mathcal{D} : G$ are *adjoint pairs* - in particular, F is a *left adjoint of G* , and G is a *right adjoint of F* - if there is a natural isomorphism

$$\text{Hom}_{\mathcal{D}}(F(X), Y) \cong \text{Hom}_{\mathcal{C}}(X, G(Y))$$

For any $X \in \mathcal{C}$ and $Y \in \mathcal{D}$.

Notation wise, it's denoted as (F, G) or $F \dashv G$.

More formally, it's a natural isomorphism $(-)^\sharp : \text{Hom}_{\mathcal{D}}(F(-), -) \xrightarrow{\sim} \text{Hom}_{\mathcal{C}}(-, G(-))$ of bifunctors $\mathcal{C}^{\text{op}} \times \mathcal{D} \rightarrow \text{Set}$ (here, we also denote $(-)^\flat$ as the inverse transformation).

This means, given morphism $f : X \rightarrow X'$ in $\mathcal{C}$, and morphism $g : Y \rightarrow Y'$ in $\mathcal{D}$, we have diagrams as follow:

$$\begin{array}{ccc} \text{Hom}_{\mathcal{D}}(F(X'), Y) & \xrightarrow{\sim} & \text{Hom}_{\mathcal{C}}(X', G(Y)) , \\ \downarrow \scriptstyle{- \circ Ff} & & \downarrow \scriptstyle{- \circ f} \\ \text{Hom}_{\mathcal{D}}(F(X), Y) & \xrightarrow{\sim} & \text{Hom}_{\mathcal{C}}(X, G(Y)) \end{array} \quad \begin{array}{ccc} \text{Hom}_{\mathcal{D}}(F(X), Y) & \xrightarrow{\sim} & \text{Hom}_{\mathcal{C}}(X, G(Y)) \\ \downarrow \scriptstyle{g \circ -} & & \downarrow \scriptstyle{Gg \circ -} \\ \text{Hom}_{\mathcal{D}}(F(X), Y') & \xrightarrow{\sim} & \text{Hom}_{\mathcal{C}}(F(X), Y') \end{array}$$

The first diagram insists that any morphism $h : F(X') \rightarrow Y$ in $\mathcal{D}$ satisfies $(h \circ Ff)^\sharp = h^\sharp \circ f$ in $\mathcal{C}$. The second diagram insists that any morphism $\ell : F(X) \rightarrow Y$ satisfies $(g \circ \ell)^\sharp = Gg \circ \ell^\sharp$. Or, the following diagram holds in $\mathcal{C}$:

$$\begin{array}{ccc} X' & & \\ f \downarrow & \searrow \scriptstyle{(h \circ f)^\sharp} & \\ X & \xrightarrow{h^\sharp} & G(Y) \end{array} \quad , \quad \begin{array}{ccc} X & \xrightarrow{\ell^\sharp} & G(Y) \\ & \searrow \scriptstyle{(g \circ \ell)^\sharp} & \downarrow \scriptstyle{Gg} \\ & & G(Y') \end{array}$$

If use the inverse transformation $(-)^\flat$, the above can also be phrased as: Any morphism $\tilde{h} : X' \rightarrow G(Y)$ satisfies $(\tilde{h} \circ f)^\flat = (\tilde{h})^\flat \circ Ff$, and any morphism $\tilde{\ell} : X \rightarrow G(Y)$ satisfies $(Gg \circ \tilde{\ell})^\flat = g \circ (\tilde{\ell})^\flat$ in $\mathcal{D}$.

A property of adjoint functor is it connects the commutativity of certain diagrams between the two categories:

Lemma 1.5. Fix arbitrary $f : X \rightarrow X'$ in $\mathcal{C}$, $g : Y \rightarrow Y'$ in $\mathcal{D}$, and two morphisms $h : F(X) \rightarrow Y$, $h' : F(X') \rightarrow Y'$ in $\mathcal{D}$. The commutativity of following two diagrams are equivalent:

$$\begin{array}{ccc} F(X) & \xrightarrow{h} & Y \\ Ff \downarrow & & \downarrow g \\ F(X') & \xrightarrow{h'} & Y' \end{array} \quad \text{in } \mathcal{D}, \quad \begin{array}{ccc} X & \xrightarrow{h^\sharp} & G(Y) \\ f \downarrow & & \downarrow Gg \\ X' & \xrightarrow{h'^\sharp} & G(Y') \end{array} \quad \text{in } \mathcal{C}$$

Proof. As a remark, $(-)^\sharp$ is a bijection on each homset, so two morphisms $F(X) \rightarrow Y'$ are equal iff after applying $(-)^\sharp$ they're equal as morphisms $X \rightarrow G(Y')$.

Using the notation from above, we see the following equivalence:

$$g \circ h = h' \circ Ff \iff (g \circ h)^\sharp = (h' \circ Ff)^\sharp \iff Gg \circ h^\sharp = h'^\sharp \circ f$$

Where, the second equivalence is the property from above. The left and right most equalities represent the two commutative diagrams respectively, hence showing the equivalence of commutativity. $\square$

Given the adjoint properties, we have the following two natural isomorphisms, given any $X \in \mathcal{C}$ and $Y \in \mathcal{D}$:

$$(-)^\sharp : \text{Hom}_{\mathcal{D}}(F(X), F(X)) \xrightarrow{\sim} \text{Hom}_{\mathcal{C}}(X, GF(X)), \quad \text{Hom}_{\mathcal{D}}(FG(Y), Y) \xleftarrow{\sim} \text{Hom}_{\mathcal{C}}(G(Y), G(Y)) : (-)^\flat$$

Hence, we can define a collection of maps $\eta_X := (\text{id}_{F(X)})^\sharp : X \rightarrow GF(X)$, and $\epsilon_Y := (\text{id}_{G(Y)})^\flat : FG(Y) \rightarrow Y$.

Proposition 1.6. *The two collections form natural transformations $\eta : \text{Id}_{\mathcal{C}} \rightarrow GF$, and $\epsilon : FG \rightarrow \text{Id}_{\mathcal{D}}$.*

Proof. Here, fix arbitrary $f : X \rightarrow X'$ in $\mathcal{C}$, and $g : Y \rightarrow Y'$ in $\mathcal{D}$. Using **Lemma 1.5**, the following two diagrams have equivalent commutativity:

$$\begin{array}{ccc} F(X) & \xrightarrow{\text{id}_{F(X)}} & F(X) \text{ in } \mathcal{D}, \\ Ff \downarrow & & \downarrow Ff \\ F(X) & \xrightarrow{\text{id}_{F(X')}} & F(X') \end{array} \quad \begin{array}{ccc} X & \xrightarrow{(\text{id}_{F(X)})^\sharp = \eta_X} & GF(X) \text{ in } \mathcal{C} \\ f \downarrow & & \downarrow GFf \\ X' & \xrightarrow{(\text{id}_{F(X')})^\sharp = \eta_{X'}} & GF(X') \end{array}$$

Where, the left side obviously commutes, so the right side commutes, showing naturality of η .

Dually, since $(\epsilon_Y)^\sharp = \text{id}_{G(Y)}$ by the inverse property, we also have the equivalence of the following two diagrams:

$$\begin{array}{ccc} FG(Y) & \xrightarrow{\epsilon_Y} & Y \text{ in } \mathcal{D}, \\ FGg \downarrow & & \downarrow g \\ FG(Y) & \xrightarrow{\epsilon_{Y'}} & Y' \end{array} \quad \begin{array}{ccc} G(Y) & \xrightarrow{(\epsilon_Y)^\sharp = \text{id}_{G(Y)}} & G(Y) \text{ in } \mathcal{C} \\ f \downarrow & & \downarrow Gg \\ G(Y') & \xrightarrow{(\epsilon_{Y'})^\sharp = \text{id}_{G(Y')}} & G(Y') \end{array}$$

Where, the right side obviously commutes, so the left side commutes, showing the naturality of ϵ . $\square$

These natural transformations are named as follow:

Definition 1.7. Given the adjoint pairs $F \dashv G$, the natural transformation $\eta : \text{Id}_{\mathcal{C}} \rightarrow GF$ is called the *unit transformation*, while $\epsilon : FG \rightarrow \text{Id}_{\mathcal{D}}$ is called the *counit transformation*.

They satisfy what's called the *Triangle Identity* as follow:

Proposition 1.8. *The following two diagrams of natural transformation holds:*

$$\begin{array}{ccc} F & \xrightarrow{F\eta} & FGF \text{ in } \mathcal{D}^{\mathcal{C}}, \\ & \searrow \text{id}_F & \downarrow \epsilon_F \\ & & F \end{array} \quad \begin{array}{ccc} G & \xrightarrow{\eta_G} & GFG \text{ in } \mathcal{C}^{\mathcal{D}} \\ & \searrow \text{id}_G & \downarrow G\epsilon \\ & & G \end{array}$$

Here, $\mathcal{D}^{\mathcal{C}}$ denotes the *functor categories* of all functors $\mathcal{C} \rightarrow \mathcal{D}$, with morphisms being natural transformations between functors.

Proof. Given arbitrary $X \in \mathcal{C}$, the goal is to show the horizontal morphisms of the following diagram are identities:

$$F(X) \xrightarrow{F\eta_X} FGF(X) \xrightarrow{\epsilon_{F(X)}} F(X)$$

Using our notation and the naturality, we see the following equality of morphisms after applying $(-)^{\sharp}$:

$$(\epsilon_{F(X)} \circ F\eta_X)^\sharp = (\epsilon_{F(X)})^\sharp \circ \eta_X = \text{id}_{GF(X)} \circ \eta_X = \eta_X : GF(X) \rightarrow X$$

But, recall that $\eta_X := (\text{id}_{F(X)})^\sharp$, using the bijectivity of $(-)^\sharp$, this enforces $\epsilon_{F(X)} \circ F\eta_X = \text{id}_{F(X)}$, showing $\epsilon_F \circ F\eta = \text{id}_F$ as natural transformations.

Dually, given arbitrary $Y \in \mathcal{D}$, the goal is again to show the following morphism is identity:

$$G(Y) \xrightarrow{\eta_{G(Y)}} GF G(Y) \xrightarrow{G\epsilon_Y} G(Y)$$

Similarly, after applying $(-)^b$ we get:

$$(G\epsilon_Y \circ \eta_{G(Y)})^\flat = \epsilon_Y \circ (\eta_{G(Y)})^\flat = \epsilon_Y \circ \text{id}_{FG(Y)} = \epsilon_Y$$

The second equality relies on the fact $(\text{id}_{F(X)})^\sharp = \eta_X$. Which, since $(\cdot)^\flat$ is again a bijection, and $\epsilon_Y := (\text{id}_{G(Y)})^\flat$, we see $G\epsilon_Y \circ \eta_{G(Y)} = \text{id}_{G(Y)}$, showing $G\epsilon \circ \eta_G = \text{id}_G$ as natural transformations. $\square$

This will be an important result when classifying monads. We also want to say the converse of Triangle Identity holds, that it generates an adjoint pair:

Lemma 1.9. *Suppose $F : \mathcal{C} \rightarrow \mathcal{D}$ and $\mathcal{C} \leftarrow \mathcal{D} : G$ are two opposite functors, together with natural transformations $\eta : \text{Id}_{\mathcal{C}} \rightarrow GF$ and $\epsilon : FG \rightarrow \text{Id}_{\mathcal{D}}$ such that Triangle Identity holds:*

$$\begin{array}{ccc}
F & \xrightarrow{F\eta} & FGF \text{ in } \mathcal{D}^C, \\
& \searrow \text{id}_F & \downarrow \epsilon_F \\
& & F
\end{array}
\qquad
\begin{array}{ccc}
G & \xrightarrow{\eta_G} & GFG \text{ in } \mathcal{C}^D \\
& \searrow \text{id}_G & \downarrow G\epsilon \\
& & F
\end{array}$$

then $F \dashv G$ is an adjoint pair.

Proof. Given any $X \in \mathcal{C}$ (associated to $\eta_X : X \rightarrow GF(X)$) and $Y \in \mathcal{D}$ (associated to $\epsilon_Y : FG(Y) \rightarrow Y$), for any morphism $h : F(X) \rightarrow Y$, we have a corresponding morphism $Gh : GF(X) \rightarrow G(Y)$. This defines the following map:

$$(-)^\sharp : \text{Hom}_{\mathcal{D}}(F(X), Y) \rightarrow \text{Hom}_{\mathcal{C}}(X, G(Y)), \quad h^\sharp := Gh \circ \eta_X$$

Similarly, for any morphism $\ell : X \rightarrow G(Y)$, we have a corresponding morphism $F\ell : F(X) \rightarrow FG(Y)$. This again defines the following map:

$$\mathrm{Hom}_{\mathcal{D}}(F(X), Y) \leftarrow \mathrm{Hom}_{\mathcal{C}}(X, G(Y)) : (-)^b, \quad \ell^b := \epsilon_Y \circ F\ell$$

These are evidently functorial. To show they're mutual inverses, we expand the following:

$$(h^\sharp)^\flat = \epsilon_Y \circ Fh^\sharp = \epsilon_Y \circ F(Gh \circ \eta_X) = \epsilon_Y \circ FGh \circ F\eta_X$$

Expanding the right hand side in terms of diagram (includes the triangle identity and naturality of ϵ), we get:

$$\begin{array}{ccccc}
& & \text{id}_{F(X)} & & \\
& \swarrow & & \searrow & \\
F(X) & \xrightarrow{F\eta_X} & FGF(X) & \xrightarrow{\epsilon_{F(X)}} & F(X) \\
& \downarrow F\eta_X & & & \downarrow h \\
& FG(Y) & \xrightarrow{\epsilon_Y} & & Y
\end{array}$$

So, we see $(h^\sharp)^\flat = \epsilon_Y \circ FGh \circ F\eta_X = h$ by the above commutative diagram.

Expand the dual diagram with the other triangle identity and naturality of η , we'll get that $(\ell^\flat)^\sharp = \ell$ instead, showing the two are mutual inverses. $\square$

1.3 Monads and Relations to Adjoint Functors

Let's first observe the case when we have an adjoint pair $F \dashv G$ from above, inducing an endofunctor $GF : \mathcal{C} \rightarrow \mathcal{C}$, with unit transformation $\eta : \text{Id}_{\mathcal{C}} \rightarrow GF$, together with *multiplication transformation* $\mu := G\epsilon_F : GF GF \rightarrow GF$ (as we have $\epsilon : FG \rightarrow \text{Id}_{\mathcal{D}}$). Notice they provide the following identities:

- This can be viewed as *Associativity of Multiplication*:

$$\begin{array}{ccc} (GF)^3 & \xrightarrow{GF\mu} & (GF)^2 \\ \mu_{GF} \downarrow & & \downarrow \mu \\ (GF)^2 & \xrightarrow{\mu} & GF \end{array}$$

Fix arbitrary $X \in \mathcal{C}$, expand the definition, we get the following diagram:

$$\begin{array}{ccc} GFG(FGF(X)) & \xrightarrow{GF G\epsilon_F(X)} & GFG(F(X)) \\ G\epsilon_F GF(X) \downarrow & & \downarrow G\epsilon_F(X) \\ G(FGF(X)) & \xrightarrow{G\epsilon_F(X)} & G(F(X)) \end{array}$$

Disregard the left most G , the above diagram commutes because $\epsilon : FG \rightarrow \text{Id}_{\mathcal{D}}$ is a natural transformation.

- This can be viewed as *Unit Property*:

$$\begin{array}{ccccc} GF & \xrightarrow{\eta_{GF}} & (GF)^2 & \xleftarrow{GF\eta} & GF \\ & \searrow \text{id}_{GF} & \downarrow \mu & \swarrow \text{id}_{GF} & \\ & & GF & & \end{array}$$

Using the definition $\mu := G\epsilon_F$, we have the following two equalities using *Triangle Identities of Adjunction*:

$$G\epsilon_F \circ \eta_{GF} = (G\psi_F \circ \eta_G)_F = (\text{id}_G)_F = \text{id}_{GF}$$

$$G\epsilon_F \circ GF\eta = G(\epsilon_F \circ F\eta) = G(\text{id}_F) = \text{id}_{GF}$$

These two identities reflect the commutativity of the above diagram.

This reflects the structure of *Monad*:

Definition 1.10. Given a category $\mathcal{C}$, a *Monad* on $\mathcal{C}$ consists of the following:

- An endofunctor $T : \mathcal{C} \rightarrow \mathcal{C}$.
- A *unit transformation* $\eta : \text{Id}_{\mathcal{C}} \rightarrow T$.
- A *multiplication transformation* $\mu : T^2 \rightarrow T$.

There are extra requirements of satisfying the following two diagrams as endofunctors on $\mathcal{C}$:

$$\begin{array}{ccc} T^3 & \xrightarrow{T\mu} & T^2 \\ \mu_T \downarrow & & \downarrow \mu \\ T^2 & \xrightarrow{\mu} & T \end{array}, \quad \begin{array}{ccccc} T & \xrightarrow{\eta_T} & T^2 & \xleftarrow{T\eta} & T \\ & \searrow \text{id}_T & \downarrow \mu & \swarrow \text{id}_T & \\ & & T & & \end{array}$$

Which are analogs to *associativity* and *unit* of monoids.

Which, starting portion shows that any adjoint pair $(F \dashv G)$ generates a monad on category $\mathcal{C}$ (the source category of the left adjoint F).

Example 1.11. As a more relevant example, let $\mathcal{T}^*$ denotes the category of based spaces, $\mathcal{T}$ denotes the category of topological spaces, there is a natural forgetful functor $F : \mathcal{T}^* \rightarrow \mathcal{T}$ that forgets the based spaces.

It has a left adjoint $(-)_* : \mathcal{T} \rightarrow \mathcal{T}^*$, where $(X)_* := X \sqcup \{*\}$ by adding a point as basepoint, and any map $f : X \rightarrow Y$ naturally extends to a based map $\tilde{f} : X_* \rightarrow Y_*$ where $\tilde{f}|_X = f$, and $\tilde{f}(*) = *$.

For any space $X \in \mathcal{T}$ and based space $Y \in \mathcal{T}^*$ (with $y_0 \in Y$ as basepoint), the natural isomorphism $\text{Hom}_{\mathcal{T}^*}(X_*, Y) \cong \text{Hom}_{\mathcal{T}}(X, Y)$ is given by $g \mapsto g|_X$, and inverse is $f \mapsto \tilde{f}$.

In this case, the unit map $\eta_X = \text{id}_{X_*}|_X$, which becomes the canonical inclusion $X \hookrightarrow X_*$; the counit map $\epsilon_Y = \text{id}_Y$, which is the map $Y_* \rightarrow Y$ that collapse the new basepoint $*$ to the original basepoint y_0 . So, the multiplication map $\eta_X = F\epsilon_{X_*} : (X_*)_* \rightarrow X_*$ by collapsing the two added basepoints (in different iteration) to the unique basepoint (with one iteration).

Since collapsing of three added points to one doesn't care about the collapse order, this explicitly forms a monad (or using the adjunction property this directly follows :)

Another question to ask: Is every monad of a category induced by adjoint pairs of functors? The answer is yes, but one needs to introduce some relevant categories constructed out of the monads.

Definition 1.12. Let (T, η, μ) be a monad on category $\mathcal{C}$. Its *Eilenberg-Moore category* for T (or the *category of T -algebras*), is a category $\mathcal{C}^T$ consists of:

- Objects of pairs (A, a) , where $A \in \mathcal{C}$ is an object, and $a : T(A) \rightarrow A$ is a morphism, so the following diagram commutes:

$$\begin{array}{ccc} A & \xrightarrow{\eta_A} & T(A) \\ & \searrow \text{id}_A & \downarrow a \\ & & A \end{array} \quad , \quad \begin{array}{ccc} T^2(A) & \xrightarrow{\mu_A} & T(A) \\ Ta \downarrow & & \downarrow a \\ T(A) & \xrightarrow{a} & A \end{array}$$

- A morphism $f : (A, a) \rightarrow (B, b)$ is a morphism $f : A \rightarrow B$, such that the following diagram commutes:

$$\begin{array}{ccc} T(A) & \xrightarrow{Tf} & T(B) \\ a \downarrow & & \downarrow b \\ A & \xrightarrow{f} & B \end{array}$$

I think intuitively $T(A)$ consists of data about “pairs of actions”, while a encodes how this action appears on A . The square provides the “associativity” of action (for instance, $(ab)\dot{v} = a(\dot{b}v)$ as action on modules). I do need to agree with professor May, that this system is more like a module instead of algebra in the traditional sense.

For this category, there is an obvious “forgetful functor” $G^T : \mathcal{C}^T \rightarrow \mathcal{C}$ by $G^T(A, a) = A$ and $G^T f = f$ for any $f : (A, a) \rightarrow (B, b)$.

Similarly, since $\mathcal{C}$ is given a monad, there is also an obvious “algebra functor” $F^T : \mathcal{C} \rightarrow \mathcal{C}^T$, by $F^T(A) := (T(A), \mu_A)$, and $F^T f = Tf$ for any $f : A \rightarrow B$. This is because μ_A satisfies the following two identities, by

the definition of monad:

$$\begin{array}{ccc}
T(A) & \xrightarrow{\eta_{T(A)}} & T^2(A) \\
& \searrow \text{id}_{T(A)} & \downarrow \mu_{T(A)} \\
& & T(A)
\end{array}
, \quad
\begin{array}{ccc}
T^2(T(A)) & \xrightarrow{\mu_{T(A)}} & T(T(A)) \\
T\mu_A \downarrow & & \downarrow \mu_A \\
T(T(A)) & \xrightarrow{\mu_A} & T(A)
\end{array}$$

Also, since $\mu : T^2 \rightarrow T$ is a natural transformation, the following diagram commutes:

$$\begin{array}{ccc}
T(T(A)) & \xrightarrow{Tf} & T(T(B)) \\
\mu_A \downarrow & & \downarrow \mu_B \\
T(A) & \xrightarrow{f} & T(B)
\end{array}$$

showing $Tf : (T(A), \mu_A) \rightarrow (T(B), \mu_B)$ is a morphism.

Here comes an important result regarding this category:

Theorem 1.13. *The two functors are adjoint pairs $(F^T \dashv G^T)$, and the induced monad is (T, η, μ) .*

Proof. Given arbitrary $A \in \mathcal{C}$ and $(B, b) \in \mathcal{C}^T$, we shall show a natural isomorphism between the two:

$$\Phi : \text{Hom}_{\mathcal{C}^T}((T(A), \mu_A), (B, b)) \xrightarrow{\sim} \text{Hom}_{\mathcal{C}}(A, B) : \Psi$$

Given a morphism $f : (T(A), \mu_A) \rightarrow (B, b)$, recall the existence of unit morphism $\eta_A : A \rightarrow T(A)$, so $\Phi(f) := f \circ \eta_A : A \rightarrow T(A) \rightarrow B$ defines a map.

Conversely, given any morphism $g : A \rightarrow B$, it naturally promotes to $Tg : (T(A), \mu_A) \rightarrow (T(B), \mu_B)$; and, since $(B, b) \in \mathcal{C}^T$, postcomposing with the morphism $b : (T(B), \mu_B) \rightarrow (B, b)$ provides $b \circ Tg : (T(A), \mu_A) \rightarrow (T(B), \mu_B) \rightarrow (B, b)$. Define $\Psi(g) := b \circ Tg$.

Remark 1.14. The reason why $b : T(B) \rightarrow B$ is also a morphism in $\mathcal{C}^T$, is because of the commutative square as follow by the fact $(B, b) \in \mathcal{C}^T$:

$$\begin{array}{ccc}
T(T(B)) & \xrightarrow{\mu_B} & T(B) \\
Tb \downarrow & & \downarrow b \\
T(B) & \xrightarrow{b} & B
\end{array}
\quad \Longrightarrow \quad
\begin{array}{ccc}
T(T(B)) & \xrightarrow{Tb} & T(B) \\
\mu_B \downarrow & & \downarrow b \\
T(B) & \xrightarrow{b} & B
\end{array}$$

The right hand side is the transpose of the left hand side, also the defining diagram of $b : (T(B), \mu_B) \rightarrow (B, b)$ as a morphism in $\mathcal{C}^T$.

The two maps Φ, Ψ are for sure functorial, hence suffices to check they're mutual inverses. Let's compute them respectively:

•

$$\Psi(\Phi(f)) = \Psi(f \circ \eta_A) = b \circ T(f \circ \eta_A) = b \circ Tf \circ T\eta_A$$

To reduce this, consider the following diagram, using triangle identities of monad and the assumptions about morphisms:

$$\begin{array}{ccccc}
T(A) & \xrightarrow{T\eta_A} & T^2(A) & \xrightarrow{Tf} & T(B) \\
& \searrow \text{id}_{T(A)} & \downarrow \mu_A & & \downarrow b \\
& & T(A) & \xrightarrow{f} & B
\end{array}$$

So, we see $b \circ Tf \circ T\eta_A = f \circ \text{id}_{T(A)} = f : T(A) \rightarrow B$ as morphism in $\mathcal{C}$ (and hence in $\mathcal{C}^T$), showing $\Psi(\Phi(f)) = f$.

•

$$\Phi(\Psi(g)) = \Phi(b \circ Tg) = b \circ Tg \circ \eta_A$$

To reduce this, again consider similar definitions (and the fact $\eta : \text{Id}_{\mathcal{C}} \rightarrow T$ is a natural transformation):

$$\begin{array}{ccccc} A & \xrightarrow{g} & B & & \\ \eta_A \downarrow & & \eta_B \downarrow & \searrow \text{id}_B & \\ T(A) & \xrightarrow{Tg} & T(B) & \xrightarrow{b} & B \end{array}$$

So, we get $b \circ Tg \circ \eta_A = \text{id}_B \circ g = g$ as morphism in $\mathcal{C}$, hence $\Phi(\Psi(g)) = g$.

This shows $(F^T \dashv G^T)$ in fact forms an adjoint pair.

To show they generate monad (T, η, μ) on $\mathcal{C}$, notice the following action on each object $A \in \mathcal{C}$ / morphism $f : A \rightarrow B$:

$$G^T F^T(A) = G^T(T(A), \mu_A) = T(A), \quad G^T F^T f = G^T(Tf) = Tf$$

Hence, $G^T F^T = T$ as endofunctor on $\mathcal{C}$.

Similarly, the unit transformation $\eta' : \text{Id}_{\mathcal{C}} \rightarrow G^T F^T$ as $\eta'_A := \Phi(\text{id}_{(T(A), \mu_A)}) = \text{id}_{T(A)} \circ \eta_A = \eta_A$, showing $\eta' = \eta$.

Finally, we have counit transformation $\epsilon' : F^T G^T \rightarrow \text{Id}_{\mathcal{C}^T}$ as $\epsilon'_{(B, b)} := \Psi(\text{id}_B) = b \circ T(\text{id}_B) = b \circ \text{id}_{T(B)} = b$. So, the induced multiplication transformation $\mu' := G^T \epsilon'_{F^T} : (G^T F^T)^2 \rightarrow G^T F^T$ satisfies $\mu'_A = G^T \epsilon'_{(T(A), \mu_A)} = G^T(\mu_A) = \mu_A$, showing $\mu' = \mu$.

Hence, $(F^T \dashv G^T)$ induces monad (T, η, μ) . □

This demonstrates each monad is induced by some adjoint pairs.

Example 1.15. For the basepoint monad demonstrated in **Example 1.11**, the Eilenber-Moore category $\mathcal{T}^{(-)*}$ consists of pairs (X, a) where $a : X_* \rightarrow X$ is a map that satisfies the following:

$$\begin{array}{ccc} X & \xrightarrow{\eta_X} & X_* \\ \text{id}_X \searrow & & \downarrow a \\ & & X \end{array} \quad , \quad \begin{array}{ccc} (X_*)_* & \xrightarrow{Ta} & X_* \\ \mu_X \downarrow & & \downarrow a \\ X_* & \xrightarrow{a} & X \end{array}$$

The first diagram enforces $a|_X = \text{id}_X$ (as $\eta_X : X \rightarrow X_*$ is the canonical inclusion $X \hookrightarrow X \sqcup \{*\}$). Given this data, the second diagram automatically commutes (by checking where the added points are sent to), so we see a is purely determined by $a(*) \in X$, in other words, it's specified by (X, x_0) for $x_0 \in X$.

Then, given a morphism $f : (X, a) \rightarrow (Y, b)$, the following diagram commutes:

$$\begin{array}{ccc} X_* & \xrightarrow{\tilde{f}} & Y_* \\ a \downarrow & & \downarrow b \\ X & \xrightarrow{f} & Y \end{array}$$

Which, $f(a(*)) = b(\tilde{f}(*)) = b(*)$ is enforced, showing f must map the chosen point in X to the chosen point in Y .

Hence, the functor $\mathcal{T}^{(-)*} \rightarrow \mathcal{T}^*$ by $(X, a) \mapsto (X, a(x))$, and $f \mapsto f$ is in fact an equivalence of category. So, the Eilenberg-Moore category of the basepoint functor is the category of based space.

Afterward, there's more question to ask: Given a monad on $\mathcal{C}$, how many ways are there to create adjoint pairs which induce the provided monad? Can we find such universal adjoint pair? For this, there are more definitions to make:

Definition 1.16. Given a monad (T, η, μ) on category $\mathcal{C}$. The *Kleisli category* $\mathcal{C}_T$ consists of:

- Objects are carried over from $\mathcal{C}$.
- Morphism $f : A \rightsquigarrow B$ in $\mathcal{C}_T$ is a morphism $f : A \rightarrow T(B)$ in $\mathcal{C}$. Under this definition, $\eta_A : A \rightsquigarrow A$ serve as the identity morphism of $A \in \mathcal{C}_T$, while given $f : A \rightsquigarrow B$ and $g : B \rightsquigarrow C$, the following morphism in $\mathcal{C}$ serves as the composition of the two:

$$g \circ_* f := A \xrightarrow{f} T(B) \xrightarrow{Tg} T(T(C)) \xrightarrow{\mu_C} T(C)$$

Proof. First, let's show associativity: Given $f : A \rightsquigarrow B$, $g : B \rightsquigarrow C$, and $h : C \rightsquigarrow D$, the two ways of composition provide the following in $\mathcal{C}$:

$$\begin{aligned} h \circ_* (g \circ_* f) &= A \xrightarrow{f} T(B) \xrightarrow{Tg} T^2(C) \xrightarrow{\mu_C} T(C) \xrightarrow{Th} T^2(D) \xrightarrow{\mu_D} T(D) \\ (h \circ_* g) \circ_* f &= A \xrightarrow{f} T(B) \xrightarrow{Tg} T^2(C) \xrightarrow{T^2h} T^3(D) \xrightarrow{T\mu_D} T^2(D) \xrightarrow{\mu_D} T(D) \end{aligned}$$

To show the two procedures agree, it suffices to show the middle right portion $\mu_D \circ Th \circ \mu_C = \mu_D \circ T\mu_D \circ T^2h$. However, this follows by the monad property:

$$\begin{array}{ccccc} T^2(C) & \xrightarrow{T^2h} & T^3(D) & \xrightarrow{T\mu_D} & T^2(D) \\ \mu_C \downarrow & & \downarrow \mu_{T(D)} & & \downarrow \mu_D \\ T(C) & \xrightarrow{Th} & T^2(D) & \xrightarrow{\mu_D} & T(D) \end{array}$$

the left square commutes by naturality of μ , the right square commutes by associativity of μ .

To show η_A serves as identity, we compute the following given $f : A \rightsquigarrow B$ and $g : Z \rightsquigarrow A$:

•

$$f \circ_* \eta_A = A \xrightarrow{\eta_A} T(A) \xrightarrow{Tf} T^2(B) \xrightarrow{\mu_B} T(B)$$

This can be simplified using triangle identity, and the fact $\eta : \text{Id}_{\mathcal{C}} \rightarrow T$ is a natural transformation:

$$\begin{array}{ccccc} A & \xrightarrow{f} & T(B) & & \\ \eta_A \downarrow & & \downarrow \eta_{T(B)} & \searrow \text{id}_{T(B)} & \\ T(A) & \xrightarrow{Tf} & T^2(B) & \xrightarrow{\mu_B} & T(B) \end{array}$$

So, we deduce $f \circ_* \eta_A = \mu_B \circ Tf \circ \eta_A = \text{id}_{T(B)} \circ f = f$.

•

$$\eta_A \circ_* g = Z \xrightarrow{g} T(A) \xrightarrow{T\eta_A} T^2(A) \xrightarrow{\mu_A} T(A)$$

Notice that $\mu_A \circ T\eta_A = \text{id}_{T(A)}$ in $\mathcal{C}$ (by triangle identity), so we see $\mu_A \circ_* T\eta_A = \mu_A \circ T\eta_A \circ g = \text{id}_{T(A)} \circ g = g$.

□

For this category, there is a functor $G_T : \mathcal{C}_T \rightarrow \mathcal{C}$, by $G_T(A) = T(A)$, and $G_T f := \mu_B \circ T f : T(A) \rightarrow T^2(B) \rightarrow T(B)$ for arbitrary $A \in \mathcal{C}_T$ and morphism $f : A \rightsquigarrow B$. This is functorial, as $G_T \eta_A = \mu_A \circ T \eta_A = \text{id}_{T(A)}$ by applying triangle identity, and given $f : A \rightsquigarrow B$, $g : B \rightsquigarrow C$, we have the following diagram commutes:

$$\begin{array}{ccccc} T(A) & \xrightarrow{Tf} & T^2(B) & \xrightarrow{T^2g} & T^3(C) & \xrightarrow{T\mu_C} & T^2(C) \\ & & \mu_B \downarrow & & \mu_{T(C)} \downarrow & & \downarrow \mu_C \\ & & T(B) & \xrightarrow{Tg} & T^2(C) & \xrightarrow{\mu_C} & T(C) \end{array}$$

This shows $G_T(g \circ_* f) = \mu_C \circ T(\mu_C \circ Tg \circ f) = \mu_C \circ T\mu_C \circ T^2g \circ Tf = (\mu_C \circ Tg) \circ (\mu_B \circ Tf) = G_Tg \circ G_Tf$.

Conversely, there is a “coalgebra functor” $F_T : \mathcal{C} \rightarrow \mathcal{C}_T$, by $F_T(A) = A$, and $F_T f := \eta_B \circ f : A \rightarrow B \rightarrow T(B)$ for all $A \in \mathcal{C}$ and $f : A \rightarrow B$ in $\mathcal{C}$. This is again functorial, as $F_T \text{id}_A := \eta_A \circ \text{id}_A = \eta_A$ is the identity in $\mathcal{C}_T$, and given $f : A \rightarrow B$, $g : B \rightarrow C$ the following diagram commutes:

$$\begin{array}{ccccc} A & \xrightarrow{f} & B & \xrightarrow{g} & C \\ & \searrow F_T f & \eta_B \downarrow & & \downarrow \eta_C \\ & & T(B) & \xrightarrow{Tg} & T(C) \\ & & \searrow T(F_T g) & & \downarrow T\eta_C \\ & & & & T^2(C) \xrightarrow{\mu_C} T(C) \end{array}$$

(Two left triangles are definition of F_T , the square is the naturality of $\eta : \text{Id}_{\mathcal{C}} \rightarrow T$, and bottom right is the triangle identity of monad).

This shows $F_T(g \circ f) = \eta_C \circ (g \circ f) = \text{id}_{T(C)} \circ \eta_C \circ g \circ f = \mu_C \circ T(F_T g) \circ F_T f = F_T g \circ_* F_T f$.

Similar to the Eilenberg-Moore category, we have the adjunction:

Theorem 1.17. *The two functors are adjoint pairs $(F_T \dashv G_T)$, and the induced monad is (T, η, μ) .*

Proof. Similar manner, we ought to show a natural isomorphism between the two homsets:

$$\Phi : \text{Hom}_{\mathcal{C}_T}(A, B) \rightleftarrows \text{Hom}_{\mathcal{C}}(A, T(B)) : \Psi$$

In fact, by definition these two sets are equal, as a morphism $f : A \rightsquigarrow B$ in $\mathcal{C}_T$, is precisely a morphism $f : A \rightarrow T(B)$, so the “identity map” between the two sets provide an obvious bijection.

About functoriality, this is established by the properties of morphisms that’re established beforehand, here we can comfortably omit it :)

It remains to check they do form the monad T : Fix $A \in \mathcal{C}$, and morphism $f : A \rightarrow B$, we get the following description:

$$G_T F_T(A) = G_T(A) = T(A), \quad G^T(F^T f) = G_T(\eta_B \circ f) = \mu_B \circ T(\eta_B \circ f) = \mu_B \circ T\eta_B \circ Tf = Tf$$

The morphism equality is due to triangle identity, $\mu_B \circ T\eta_B = \text{id}_{T(B)}$. This shows $T = G_T F_T$.

About the unit transformation $\eta' : \text{Id}_{\mathcal{C}} \rightarrow G_T F_T$ is given by $\eta'_A = \Phi(\eta_A) = \eta_A$ (because η_A is the identity of A in $\mathcal{C}_T$), showing $\eta' = \eta$.

About the counit transformation $\epsilon' : F_T G_T \rightarrow \text{Id}_{\mathcal{C}_T}$, every $B \in \mathcal{C}_T$ satisfies $\epsilon'_B = \Psi(\text{id}_{T(B)}) = \text{id}_{T(B)} : T(B) \rightsquigarrow B$. Then, the multiplication transformation $\mu' := G_T \epsilon_{F_T}$ satisfies $\mu'_A = G_T \epsilon_{F_T(A)} = G_T(\text{id}_{T(A)}) = \mu_A \circ T\text{id}_{T(A)} = \mu_A \circ \text{id}_{T^2(A)} = \mu_A$, showing $\mu' = \mu$.

So, the given adjunction does induce the monad (T, η, μ) , finishing the proof. $\square$

Finally, the universality of these two categories are given as follow:

Theorem 1.18. *In the adjoint category Adg_T (categories of all category with adjoint functors between $\mathcal{C}$, inducing monad T ; the morphisms are functors that commutes with the left/right adjoints for each category), the Kleisli category $(\mathcal{C}_T, F_T \dashv G_T)$ is initial, while Eilenberg-Moore category $(\mathcal{C}^T, F^T \dashv G^T)$ is final.*

Lemma 1.19. *The canonical functor $\mathcal{C}_T \rightarrow \mathcal{C}^T$ is fully faithful, with essential image being all the free T -algebra (all pairs $(T(A), \mu_A)$).*

1.4 Monad Algebra and Monadicity